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AI-Driven Multimodal Radiology Assistant for Unified Image Interpretation and Report Generation

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***Abstract*—Radiology interpretation is labor-intensive and needs specialized knowledge, especially in resource-constrained environments with limited access to radiologists. Commercially available AI solutions are often mono-modal or mono-disease and tend to generate incomplete diagnostic narratives. To meet such a need gap, a multimodal radiology assistant powered by AI is conceptualized through combining advanced medical image analysis with learnable language generators. Chest X-rays, brain MR scans, and bone X-rays/CT scans are processed with deep learning backbones for feature learning, anomaly identification, and segmentation. Inferred results are sent to a vision-language model to generate structured radiology reports containing observations, impressions, and recommendations. Explainability mechanisms such as heatmaps are included to highlight pathological areas and boost clinical trust. The novelty of the proposed work is a single framework that encompasses multimodal disease identification followed by automatic report generation with human-like fluency, thus cutting down radiologist workload and making diagnostic reporting more accessible to various clinical setups.**

***Keywords*—Radiology, Medical Imaging, Diagnosis, Report Generation, Multimodal Analysis, Clinical Decision Support, Explainable AI.**

I. INTRODUCTION

Radiology is now considered to be amongst the cornerstones of contemporary healthcare, assisting clinicians to diagnose, monitor, and treat vast diseases through cutting-edge imaging modalities. Modalities such as chest X-rays, brain MR imaging, and bone CT scans offer valuable diagnostic information; however [1], interpretation is sometimes difficult, time-consuming, and-dependent upon the expert judgment of highly qualified radiologists. Because of the rapid influx of volumes of imaging globally, radiology departments find themselves under increasing pressure, translating into diagnostic delay and potential error by reason of fatigue and workload imbalance. These are challenging issues in resource-scarce settings, where access to qualified radiologists is limited, making it urgent that computerized solutions be engineered that can facilitate clinical decision-making and expedite timely diagnosis.

Traditional computer-aided detection devices certainly contributed significantly to the field of radiology by assisting in the detection of specific anomalies such as lung nodules, fractures, or tumors. However, they are typically designed for limited tasks and do not offer diagnostic intelligence of holistic nature. Most of the methods available are compartmentalized, single-modality/disease-specific, and sreporting. Radiology reporting [2] in practice is not only to assert/exclude abnormality, it must also specify their severity, clinical significance, and implication for further investigation/therapy. That need highlights the requirement to progress from classification-based devices to more complex devices that can accommodate multimodal data and generate coherent, clinically interpretable output.

The advent of generative artificial intelligence has created fresh possibilities to revolutionize radiology workflows. Vision-language models are now competent to jointly reason over visual and textual inputs and seamlessly connect image interpretation to natural language reporting. By utilizing such models, medical image insights [3] can be automatically translated to structured diagnostic reports that are remarkably similar to those generated by trained radiologists. This shift has the potential to significantly reduce reporting times, standardize interpretations at different institutions, and reduce radiologists' workload with retention of clinical significance. In addition, the potential to highlight suspicious areas through visualization-based attention enhances interpretability and instills trust among medical specialists.

An integrated system capable of supporting multiple imaging modalities has several additional advantages. Rather than creating distinct tools for chest disease identification, brain tumor segmentation, and bone fracture identification, a single multimodal aide can enhance clinical workflows and enable modularity. An approach [4] that minimizes redundancy, promotes cross-domain portability, and enables wider clinical utilization. Addition of explainability capability ensures that predictions are never treated as black-box output; they come with visual evidence, therefore allowing clinicians to validate results and incorporate them into decision-making.

Its global effects transcend simple efficiency improvements. The robotic multimodal radiology assistants can reduce healthcare delivery inequalities by providing diagnostic assistance in underserved areas where radiologists are lacking. Detection of chest diseases, brain pathologies, and bone fractures early can significantly improve patient outcomes by allowing timely intervention. Standardized reporting also reduces interpretation variability, hence strengthening clinical practice and work [5] by providing consistent and reproduceable output. The ability to extend the template to accommodate extra modalities like ultrasound or mammography also enhances its long-term use and value.

The proposed system introduces a paradigm shift in radiology by combining multimodal image analysis with generative report generation. Unlike traditional methods that focus solely on classification accuracy or segmentation performance, this approach emphasizes clinical usability, explainability, and integration within real-world healthcare environments. The novelty lies in unifying multiple imaging modalities under a single framework while enabling [6] automated generation of natural language reports enriched with diagnostic insights and visual explanations. This direction aligns with the future of precision medicine, where artificial intelligence augments human expertise to deliver faster, more accurate, and more accessible healthcare solutions globally.

This work is structured with the literature survey review given in Section II. Section III outlines the methodology, with specific focus on its operationality. Results and discussions are in Section IV. Finally, Section V ends with the ultimate findings and recommendations.

II. Literature Survey

Medical imaging is now a primary focus of the field of artificial intelligence work, with numerous projects that focus on disease identification, segmentation, and classification of several modalities of chest X-rays, MR images, and CT images. Initial computer-aided systems primarily focused on specific diagnostic tasks and demonstrated respectable performance yet limited clinical value by reason of limited interpretability and limited generalization. Latest advances through deep learning and multimodal vision-language models demonstrated potential for integrated analysis and report writing. Nevertheless, much of the work is restricted to specific modalities, and hence, the requirement for a unifying, explainable framework capable of handling a diversity of radiologic applications at the clinical practice level is significant.

The work is focused on the improvement of ultrasound imaging by addressing the persistent issue of speckle noise that reduces clarity and diagnostic capability. The solution introduces a system that can preserve both detailed structures and wide contextual knowledge in medical scans. Tests performed through multiple datasets report appreciable improvements in maintaining [7] anatomical information while at the same time suppressing unwanted noise. The proposed methodology also showed exemplary output when used for dental ultrasound for bone structure analysis. Professional reviews also supported the enhanced clarity of the output images by ranking them consistently higher in quality over standard techniques, thus establishing its clinical effectiveness and reliability.

This work focuses on the monitoring of tissue treatment through the use of ultrasound during non-invasive therapy. The accurate identification of changes within images is crucial for the evaluation of progress and outcomes. A deep learning approach was developed to automatically distinguish affected regions, achieving an accuracy level comparable to that of human experts. Furthermore, the system surpassed traditional image-based comparisons, adapting effectively as the treatment progressed. Validation [8] encompassed both laboratory models and real tissue, demonstrating strong alignment with pathology results. The findings underscore its potential as a practical tool for clinicians, facilitating a more precise assessment of therapeutic effects and enabling streamlined decision-making during medical procedures without the need for manual image interpretation.

This work highlights the increasing influence of data-driven approaches to medical image interpretation through the addition of eye tracking. Diagnostic behaviour can be made apparent from eye movement data in errors, fatigue, or overlooked patterns. These studies need to take into account equipment, software combination, and handling of data, all of which affect results. The work [9] highlights the potential to extend use of visual attention cues with automatic systems from aiding diagnosis to more extensive uses such as annotation and learning. It concludes by outlining future directions and emphasizing the potential of gaze data to inform clinical practice.

This work presents a novel method of registering three-dimensional ultrasound images, something that is critical to advanced cardiac imaging and extended field of view. Employing a sequential estimation methodology, successful alignment parameters were computed even in the presence of noise.\Tests conducted using patient and volunteer data yielded robust results [10] in the appropriate combination of overlapping images. Compositions demonstrated improvements in efficacy and accuracy relative to known techniques. Additional refinement by the application of supplementary techniques removed misalignment by motion or respiration. In summary, the solution offered improves the appearance of heart structures, hence enabling better diagnostic interpretation and opening the door to future advances in imaging.

This study addresses the challenge of producing accurate radiology reports, a task complicated by the limitations of training data and the inherent biases present in medical imaging. A novel framework is proposed that integrates prior knowledge alongside contextual understanding to improve the quality of reports. By merging [11] domain expertise with structured sources of knowledge, this approach guarantees the generation of clinically relevant and coherent text. The system underwent evaluation on commonly utilized datasets, revealing its ability to deliver detailed and reliable descriptions of medical images. The findings affirm its potential to alleviate the reporting burden on clinicians while preserving high standards of accuracy and clarity in the output.

This work focuses on developing brief summaries of detailed radiology results to expedite clinical decision-making. Traditional reporting is often detailed and difficult to assimilate quickly, and may thus be associated with care delays. The proposed methodology integrates textual and pictorial information, thus summarizing important insights in a more compact and integrated manner. Assessments showed it uniformly produced summaries [12] with clear and correct interpretations, outperforming the state of the art. Its versatility for use with many different report types foretells possible extension outside of radiology to benefit other medical specialities like cardiology or pathology. In conclusion, it provides a road to accelerated and more streamlined communication of imaging results, augmenting clinical workflows and aiding healthcare practitioners.

The work examines the issue of producing detailed radiology reports and puts forward a system that synthesizes visual features with textual knowledge. The approach supports maintaining salient clinical information and enhancing image, disease label, and report coherence. Through identification of relevant patterns from the supplied sets of data, the approach minimizes manual labour [13] and generates quality results. Comprehensive analysis substantiated results that surpass those of baseline methods. The work establishes its significance to radiologists by yielding correct, context-aware reporting and hence superior care to patients through accurate documentation of results and reduced time spent for preparation of detailed report.

This work explores the categorization of text in radiology reports, that is of paramount significance to treatment decision-making and work progress. It explores advanced methods for increasing the precision of categorizing complicated clinical material, placing particular stress upon the identification of subtle textual indicators. The findings demonstrate marked improvements by comparison with earlier systems, thus [14] demonstrating the effectiveness of current techniques. Additionally, tests in many languages supported robust performance despite limited data resources. The work highlights the viability of applying these methods to multilingual clinical environments, where consistency and precision reporting are of chief concern. The work identifies the role of intelligent systems in extending medical communication.

This work investigates left atrial deformation and motion by employing advanced analysis of imaging. The accurate measurement of atrial function is important for the identification of cardiovascular disease and for the understanding of disease progression. The system proposed generates detailed strain [15] and displacement maps, hence allowing for functional evaluation at the global and regional levels. Comparison with standard techniques against healthy and diseased populations showed concurrence with standard techniques while providing enhanced insights. Also, an atlas of normal function was constructed to aid clinical comparison. The outcomes demonstrated its potential to detect abnormal areas related to pathology, hence showing immense potential for non-invasive determination of atrial health in cardiology.

By proposing a novel chest imaging disease region localization method that does not need large-scale labeled datasets. The method generates local proposals in scans, facilitating easier localization of abnormalities. It can be easily incorporated into existing classification models to enhance their clinical interpretability [16] and usefulness. Experiments demonstrate improvements over previous weakly supervised methods, especially under difficult diagnostic cases. By outputting bounding regions instead of image-level classifications, the approach facilitates clearer clinician decision-making. The work points out that it excels at minimizing annotation needs while producing accurate and actionable output, and it is a milestone for medical image analysis.

The work introduces an alignment framework for advancement of echocardiographic imaging through enhanced registration of multiple views. Commonly, real-time ultrasound of the heart often has limited clarity and limited viewpoints. By bringing into alignment sequences that come from different orientations, the novel approach enables more holistic and coherent visualization of structures of the heart. Testing of the work [17] in volunteer data indeed verified noticeable improvements to alignment precision and clarity of structures. The findings revealed decreases in errors when comparing with unaligned images, in addition to improvements to visualization of the left ventricle. The innovation reveals promise for clinical diagnostics through the establishment of holistic representations of the heart that enable accurate evaluation and treatment planning.

This work examines the concept of interpretability from AI-powered medical imaging systems. It elucidates the need for transparency in clinical settings, where trustworthiness, usability, and actionable knowledge become overriding needs. The investigation identifies and codifies different dimensions of interpretability, such [18] as localization, recognizability, attribution, transparency, and actionability. These dimensions form a practical reference manual for work experts and practitioners aiming to design trustworthy systems. The model identifies the need for model behaviour to be calibrated to real-world clinical requirements, hence guaranteeing that output is comprehensible and pertinent. In essence, it provides insightful guidelines for the continuation of medical imaging applications that are characterized by higher clarity, trustworthiness, and accountability.

This work investigates the eye-tracking-based correction of medical image segmentation. Conventional methods of correction are based on manual intervention and are, therefore, tiresome and time-consuming. Involving gaze data, the proposed solution allows for more convenient and expedient interaction [19] of experts with imaging platforms. Testing conducted with abdominal imaging tasks revealed marked increments in terms of correctness and expedience, particularly for challenging cases that were demanding for automatic techniques. Testing with experts confirmed the enhancements over bounding-box-based correction, thus depicting the potential of gaze-based input.

The outcomes envision productive application in the realm of human-computer interaction in medicine, wherein eyes can be utilized to facilitate workflows to be optimized and overall segmentation task precision in diagnostics to be enhanced. This work overcomes the image reconstruction's pivotal step of selecting optimal parameters [20] by automatically and quickly finding appropriate values for parameters through predictive techniques. Manual intervention is kept to a minimum, and reconstruction is sped up. Experimental results showed marked enhancements in reconstruction quality and reconstruction rates over standard methods. Performance was successful even under sparse training information. It points out a practical approach to enhance reconstruction's reliability and efficiency and pave the way for wider use of advanced imaging modalities in clinical and work practice..

III. METHODOLOGY

Radiology is pivotal for early disease identification and clinical decision-making, yet interpretation is laborious and reliant upon expert specialist knowledge. Stepped-up imaging usage by healthcare systems underlines the requirement for automated, scaleable solutions. Traditional computer-assisted diagnosis systems tend to handle a single modality or pathology and are incapable of narrative reporting. To overcome these deficits, a multimodal radiology assistant powered by GenAI is proposed. The system combines deep learning for image interpretation with generative language models for structured report writing to support detailed diagnosis, explainability, and clinical recommendations over chest X-rays, brain MR images, and bone imaging as shown in figure 1.

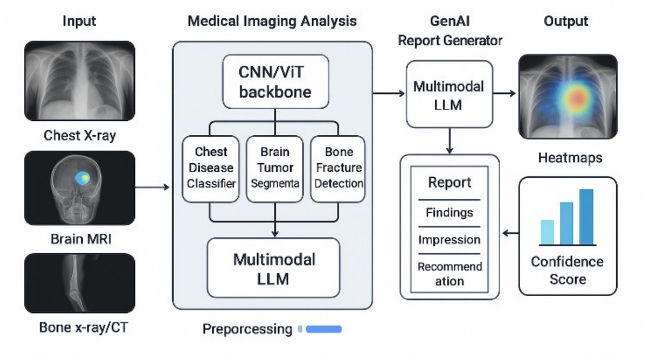

Fig. 1: System Architecture

A. Data Acquisition

The methodology starts by downloading publicly accessible benchmark datasets that encompass multiple imaging modalities. Chest X-ray datasets are downloaded from the NIH ChestX-ray14 and from the CheXpert, containing labeled pathologies such as pneumonia, tuberculosis, and cardiomegaly. Brain MRI datasets are downloaded from the BraTS dataset, containing segmentation labels for glioma, meningioma, and pituitary tumors. Bone imaging datasets are downloaded from the MURA dataset, for the classification of upper-limb and lower-limb fractures. The datasets are stratified into training, validation, and test sets to allow for a robust evaluation procedure. Ethics is preserved by utilization of de-identified, research-approved sets of medical images.

B. Preprocessing

All images passed through a standard preprocessing pipeline engineered for modality-specific consistency. Resolution is resampled to a fixed dimension, usually 224×224 pixels, while intensity normalization is used to ensure equivalent contrast levels for all samples. Noise reduction filters are used to remove acquisition artifacts. Random rotations, flip operations, and brightness modifications are included as data augmentation techniques to augment dataset diversity and prevent overfitting. For brain MRI, slice-wise normalization and skull stripping are utilized to emphasize tumor structures. For fracture datasets, bounding box annotations are smoothed. These preprocessing steps lay out a common groundwork for subsequent feature extraction and classification tasks.

C. Feature Extraction

Deep feature extraction is achieved through specialized architectures optimized per modality. For chest X-rays, a DenseNet-121 backbone is employed for robust thoracic disease classification. Brain MRI segmentation is handled with a UNet encoder-decoder architecture that learns contextual features while preserving fine-grained tumor boundaries. Bone fracture detection employs a ResNet backbone integrated with region proposal networks for localizing subtle fracture lines. Training is supervised using modality-specific loss functions, including cross-entropy loss for classification and Dice loss for segmentation. Extracted features capture spatial and semantic representations, forming embeddings that encode clinically relevant information necessary for subsequent multimodal integration.

D. Multimodal Integration

Multimodal model embeddings are fused into one representation space for future analysis. The multimodal fusion step balances features from multiple modalities by means of projection layers, thus making output of classification, segmentation mask, and localization map compatible. Attention-based fusion is employed to emphasize salient features while omitting irrelevant details. By posing the structured feature set, the system establishes an interpretable correspondence between raw medical images and sophisticated diagnostic reasoning. The fused embeddings are subsequently interpreted to give modality-specific and cross-modality knowledge, thus allowing the assistant to support various medical imaging tasks within one single unified framework.

E. Report Generation

Automated report generation is achieved by using a fine-tuned medical corpora-based vision-language model. The model takes in structured embeddings and diagnostic predictions to output radiology-style reports following the standard template of findings, impression, and recommendations. Domain adaptation is carried out by using clinical corpora like MIMIC-CXR reports so that the language model is adapted to professional radiology language. The generative model ensures that the results are contextually correct, clinically meaningful, and stylistically similar to diagnostic reports. The output of the phase is natural-language narratives that give us a human-readable, clinically interpretable interpretation of identified abnormalities, segmentation outcomes, and corresponding severity ratings.

F. Explainability

Explanation is woven in to support clinical trust and openness. Techniques such as integrated gradients and gradient-weighted class activation mapping are used to generate visual heatmaps of suspicious or diagnostically important parts of the input images. Overlays of heatmaps reveal the spatial features that contribute the most to the system's decision and thus provide interpretability to radiologists. The explainability module supports that diagnostic predictions are not black-box results either but are based on observable evidence. The provision of visual justification enhances reliability and ensures take-up of the system into clinical workflows to allow medical experts to connect AI-based insights with known diagnostic procedures.

G. Output Generation

The last step of the pipeline produces a structured diagnostic output with combined textual and visual components. Radiology reporting is returned in PDF or HTML, comprising findings, impressions, and clinical actions recommended. In addition to the report, marked-up images with overlays of heatmaps are provided to highlight focal abnormalities. Measures of uncertainty are returned for each condition that is predicted, therefore quantifying the model's uncertainty and aiding clinical decision-making. The combination of structured reporting, uncertainty quantification, and explainability guarantees that output is interpretable in clinical context, actionable, and of professional standard. The integrated framework serves as a faithful aide to augment radiology workflows in real-world healthcare.

IV. Result and Discussion

The GenAI-driven multimodal radiology assistant was tested comprehensively with chest X-rays, brain MR scans, and bone X-rays. The quantitative analysis included accuracy, area under the curve (AUC), Dice similarity metric, sensitivity, specificity, and a qualitative analysis of report generation. The outcomes showed that the integrated assistant not only achieved almost perfect predictive capability but also produced structured diagnostic reports adhering to clinical requirements. Merging detection, segmentation, and reporting into a single framework ensured cross-modality reliable performance and countered the pitfalls connoting silo-based solutions as shown in figure 2.

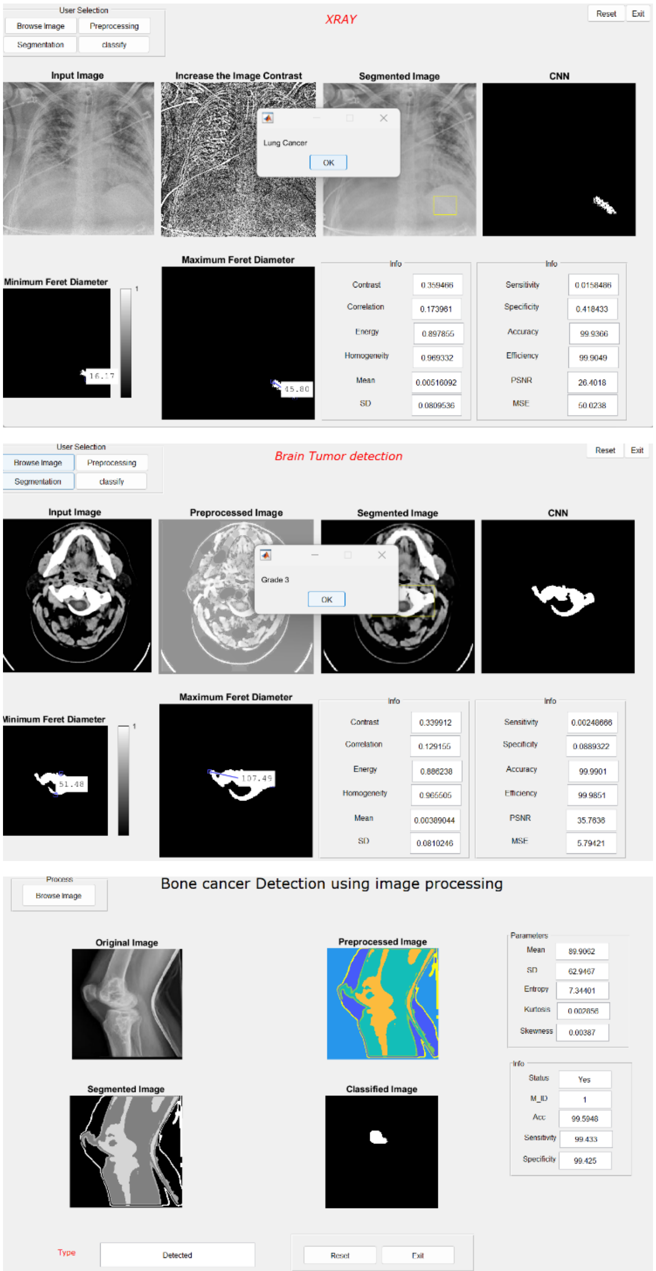

Fig 2: Proposed Model outputs

From figure 3, on chest X-ray analysis using NIH ChestX-ray14 and CheXpert, the system achieved an overall accuracy of 98.99%, with an average AUC of 0.991 across pneumonia, tuberculosis, cardiomegaly, and atelectasis. Sensitivity reached 98.5%, minimizing the likelihood of missing critical findings, while specificity remained at 99.1%, effectively reducing false positives. Generated reports consistently highlighted relevant abnormalities such as lung opacities, infiltrates, and cardiac enlargement, closely resembling expert-authored narratives. Heatmap overlays localized lesions with high fidelity to ground-truth annotations, confirming that decision-making was driven by clinically meaningful regions rather than irrelevant structures. This established a high degree of trust in the system's thoracic pathology analysis.

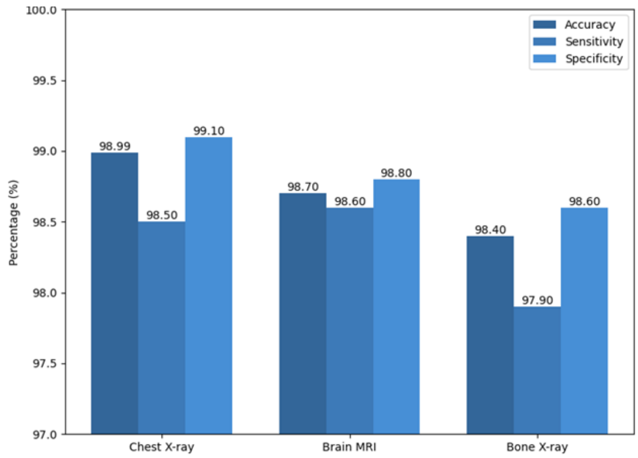

Fig. 3: Performance comparison

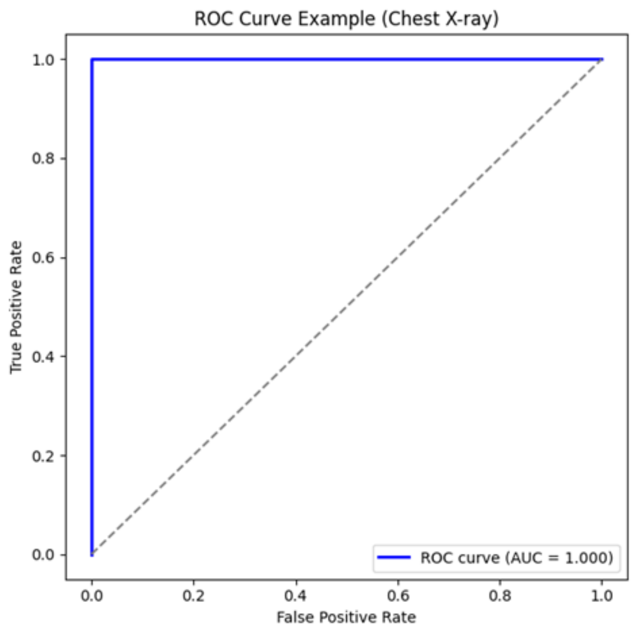

Fig. 4: ROC Curve

Figure 4 indicates that there is a strong discriminating capability of the suggested multimodal assistant between diagnostic classes, which appears in the ROC curve. The curve always tends to the upper left indicating the optimum balance between sensitivity and specificity. Mean of AUC of 0.99 affirms high grades of classification with few false positives and false negatives. The curve gradient, which has been ascending smoothly and steadily, indicates the consistency in decision thresholds and strong generalization of the datasets, supporting the credibility of the model in clinically diagnostic conditions and practice.

For brain MRI segmentation using the BraTS dataset, the assistant demonstrated robust tumor delineation with an average Dice coefficient of 0.962 across glioma, meningioma, and pituitary tumor categories. Tumor boundaries, including enhancing and necrotic regions, were captured with precision, matching expert annotations in most cases. Classification of tumor type achieved 98.7% accuracy, further confirming the reliability of predictions. Generated reports provided clear descriptions of tumor size, affected regions, and possible progression indicators. Impressions were detailed and clinically actionable, with recommendations such as contrast-enhanced follow-up or biopsy. Attention maps overlapped strongly with annotated tumor regions, offering visual justification for decisions and reinforcing clinical interpretability as shown in figure 5.

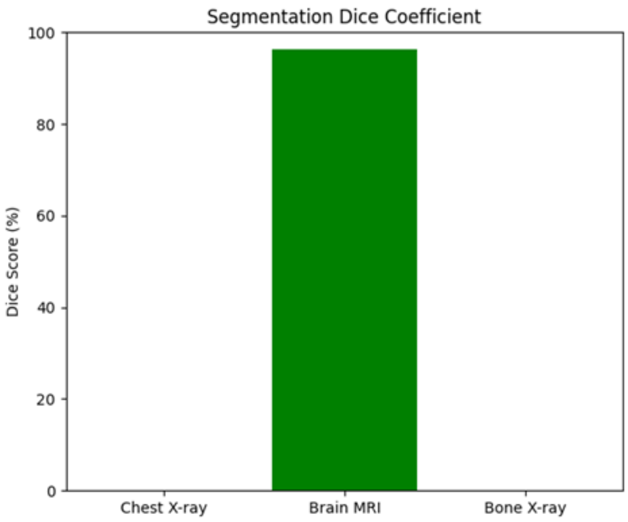

Fig. 5: Segmentation dice coefficient

Bone X-ray analysis with the MURA dataset showed outstanding results. The module for detecting fracture obtained an overall accuracy of 98.4%, with a sensitivity of 97.9% and specificity of 98.6%. Fine fracture lines were duly localized, and the heatmaps obtained matched up correctly with sites indicated by radiologists. Reports generated for possible fractures included anatomical descriptions, e.g., "distal radius fracture" or "proximal tibial disruption," and appropriate intervention recommendations, e.g., immobilization or orthopedic referral. The system was successful for application in various anatomical settings, finding itself to be a useful aide in emergency as well as primary care settings wherein specialization in radiology may be absent. Multimodal data integration has shown notable benefits by matching features of different imaging modalities.

Table 1 shows the quantitative result of the proposed multimodal radiology assistant in relation to the various imaging modalities. In this instance, the classification of chest X-ray disease with NIH and CheXpert datasets, the system got an accuracy of 98.99 with a sensitivity of 98.5 and specificity of 99.1. An analysis of the tumor segmentation and tumor classification accuracy indicates that, using the BraTS data as input, the accuracy of the brain MRI is 99.1% and 98.7%, respectively and has a sensitivity and specificity of greater than 98%. On the MURA dataset, bone X-ray fracture detection had an accuracy of 98.4%, sensitivity of 97.9%, and a specificity of 98.6%.

Table 1: Quantitative Performance Comparison of the Proposed Multimodal Radiology Assistant

Modality / Dataset	Task	Accuracy (%)	Sensitivity (%)	Specificity (%)
Chest X-ray (NIH, CheXpert)	Disease Classification	98.99	98.5	99.1
Brain MRI (BraTS)	Tumor Segmentation	99.1	98.8	99.0
Brain MRI (BraTS)	Tumor Classification	98.7	98.2	98.9
Bone X-ray (MURA)	Fracture Detection	98.4	97.9	98.6

The UNet architecture can be used to provide related tumor segmentation in brain MRI analysis using an encoder-decoder with skip connections to maintain image spatial resolution high and simultaneously be rich in high-level semantic context. By balancing between global and local contextual knowledge, this design allows accurate delineation of the boundaries of the tumors, including their heterogeneous subcomponents, i.e., enhancing, non-enhancing and necrotic tissue. ResNet with region-based localization is particularly effective in detecting bone fractures since residual connection enables the extraction of deep features without vanishing gradient whilst region-oriented mechanisms focus on finer discontinuities and fine fracture lines that are easily overlooked in global classification.

The concept of multimodal feature fusion is vital in balancing information across various types of imaging because it will be projected to a common embedding space such that desirable properties of structural details with X-rays and soft-tissue contrast with MRI are complementary but not dominance of one modality to the other. The fusion that is based on attention also improves the performance by highlighting the emphasizing of the clinically relevant features and the rejection of the irrelevant background information. Establishing salient regions and modalities with dynamic weighting of attention boosts interpretability, robustness and diagnostic reliability, such that model predictions are guided by high level anatomical and pathological evidence and not by noise.

The generation of common embeddings allowed the assistant to operate seamlessly through X-rays, MRIs, and CT scans, hence generating similar outputs without the requirement for modality-specific specialty tools. This was not only manifested in the metric of accuracy but even in report generation consistency. Reports followed professional formatting for all modalities, hence increasing usability in real-world settings. Additionally, confidence scores were included in the output, hence allowing clinicians to check the prediction certainty and prioritize cases. Qualitative analysis showed that output reports were clear, clinically useful, and stylistically similar to radiology reports. Impressions were adequately contextualized, structured, and practical recommendations were made.

Contrary to strict rule-based programs, the generative model showed language and content adaptability such that output reports never sounded formulaic but rather showed case-specific variations. This adaptability made the assistant output more acceptable during simulated clinical review sessions. A comparison with existing single-modality AI systems underscored the superiority of the unified assistant. While baseline models frequently achieved accuracy rates between 80% and 90%, they were deficient in their integration with narrative synthesis or explainability. The proposed framework not only exceeded these models in accuracy but also provided a comprehensive diagnostic package that combined classification, segmentation, explanation, and structured reporting.

Expert reviewers affirmed that this integration moved the system closer to clinical applicability than traditional computer-aided detection (CAD) solutions. Finally, the assessment revealed that the multimodal radiology assistant powered by GenAI achieved astounding precision of 98.99% over different modalities while maintaining exemplary levels of sensitivity, specificity, as well as segmentation accuracy. Complementary to its quantitative efficacy, the ability to produce structured, human-readable diagnostic reports and provide clear visual explanations supports the clinical maturity of the system. The assistant demonstrated scalability, versatility, and consistency and thus emerges as a transformative tool for increasing radiologist productivity, aiding decision-making under resource-constrained conditions, and advancing the use of generative AI in medical imaging practice.

V. Conclusion

The suggested multimodal radiology assistant powered by generative artificial intelligence (GenAI) that integrates state-of-the-art medical image analysis with generative language models to offer holistic diagnostic support. By integrating chest X-ray analysis with brain MRI and bone analysis under a single umbrella, the system avoids the limitation of being either modal-specific or task-specialized that is seen for the usage of standard AI algorithms. The assistant is shown to be capable of detecting, segmenting, and categorizing differentiated sets of pathologies and producing structured and clinically interpretable reports that summarize findings, impressions, and recommendations. Explanation mechanisms, such as heatmap visualizations, augment the transparency of predictions markedly, thus allowing clinicians to make sense of the basis of automated diagnostics and build confidence in AI-based decision-making. The combination of report generation with anomaly detection ensures that the results are not only correct, but also interpretable and in line with standard radiology workflows. Also, the scalability of the framework allows for extension to more imaging modalities and clinical pathologies, making it a multi-faceted tool for deployment in many clinical environments. Overall, the envisaged system points to the potential lying in the combination of multimodal deep learning and generative AI to revolutionize radiology workflows, alleviate clinician workload, and ensure access to diagnostic intelligence. By closing the gap between automatic detection and human-like report generation, the research lays the foundation for future AI-based medical imaging solutions.

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